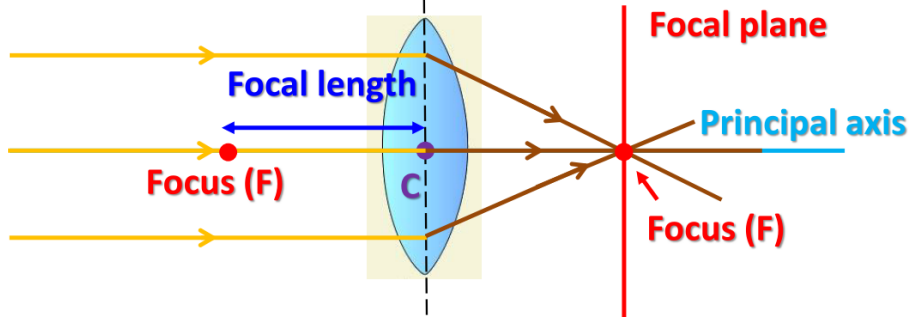
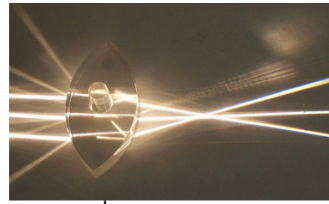


Convex lens



incident rays **parallel**
to **principal axis**

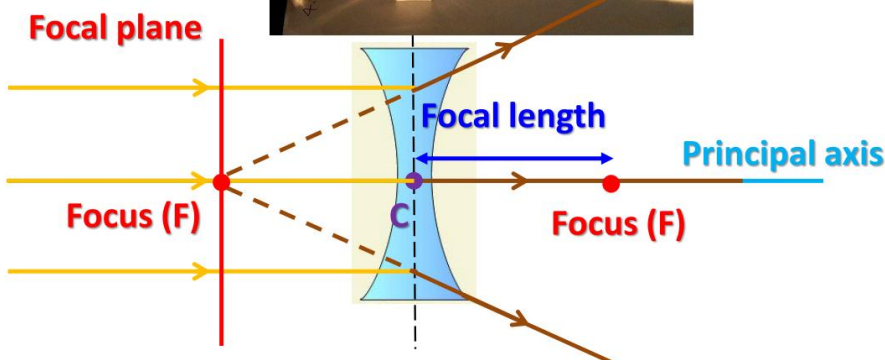
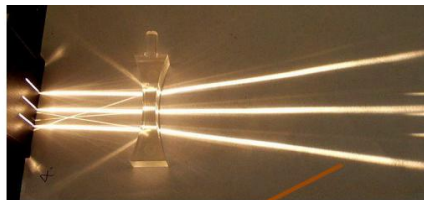
Refracted rays **converge**
to a **point (Focus)**

Convex lens = Converging lens

Symbol for
convex lens



Concave lens



incident rays **parallel**
to **principal axis**

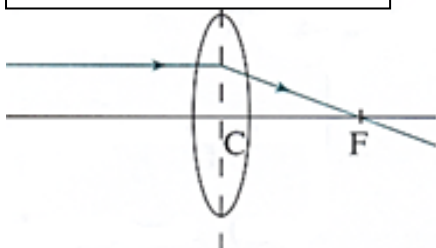
Refracted rays **diverge**
from a **point (Focus)**

Concave lens = Diverging lens

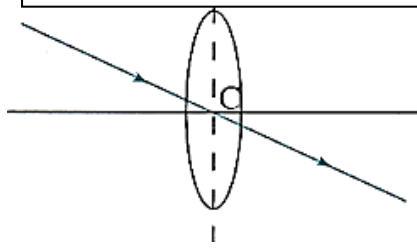
Symbol for
concave lens



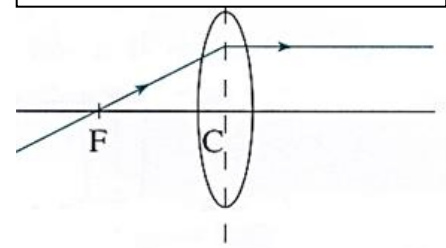
平衡入, Focus(F) 出



C 入 C 出, 直入直出

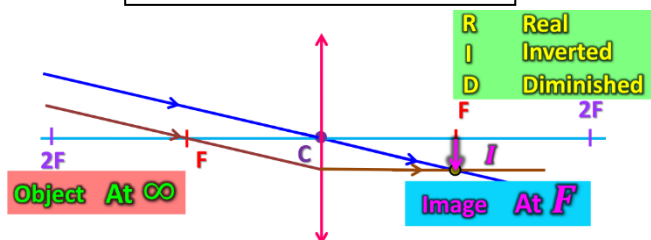


Focus(F) 入, 平衡出

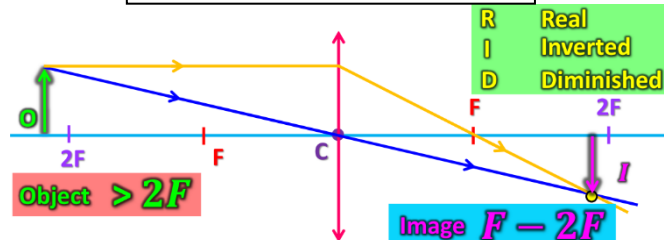


Construction of images by ray diagrams (i) Convex lens

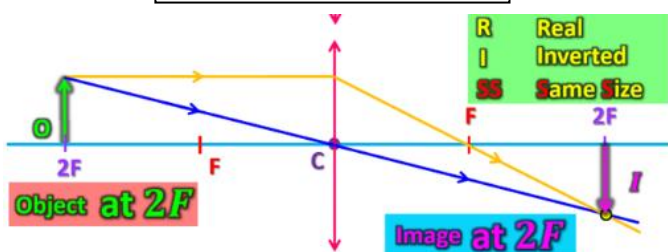
1. Object placed at infinity



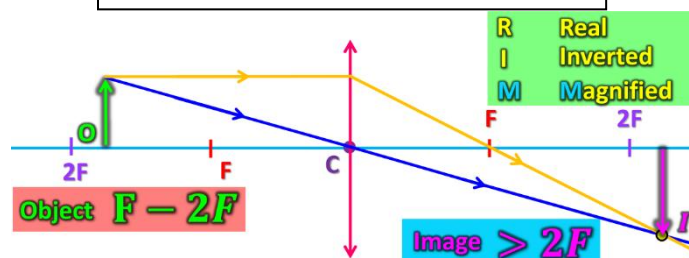
2. Object placed beyond $2F$



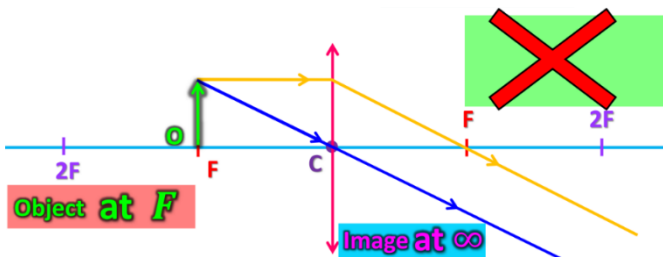
3. Object placed at $2F$



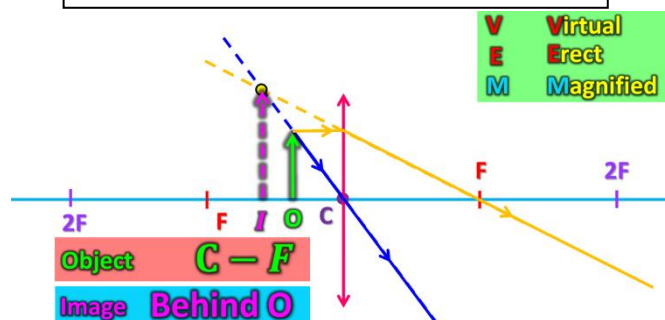
4. Object placed between F and $2F$



5. Object placed at focus F



6. Object placed between lens and F



Remarks:

- Convex lens can only give same size image when the object is placed at $2F$. Separation between the object and the image is then equal to 4 times of focal length, which is the minimum separation between object and the real image.
- Light is reversible and hence the position of the object and the real image can be interchanged.
- A lens with a larger curvature or greater refractive index bends light more and hence it has shorter focal length
- When part of lens is covered, a **complete** but **dimmer** (lower brightness) image is formed since some light rays are blocked and amount of light forming the image decrease.
- When two convex lenses of the **same size but different focal length f** are placed at the same distance from an object, same amount of light is distributed over the image in each case so that **smaller image** is **brighter** but **larger image** is **dimmer**.
- **Application** of convex lens:
 - (i) Magnifying glass
 - (ii) Spectacles for correcting long-sighted eye

1. Which of the following optical devices can form a virtual and magnified image?
- (1) Convex lens
 - (2) Concave lens
 - (3) Plane mirror
- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only
2. Which of the following optical devices can form a real and magnified image?
- (1) Concave lens
 - (2) Convex lens
 - (3) Plane mirror
- A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only
3. Parallel rays from a distant object converge on passing through a lens. Which of the following are correct?
- (1) The lens is convex.
 - (2) The image formed is real.
 - (3) Image distance is equal to focal length.
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)
4. Which of the following descriptions about a convex lens is/are correct?
- (1) A convex lens is a converging lens.
 - (2) The image formed by a convex lens is always real.
 - (3) The image formed by a convex lens is always magnified.
- A. (1) only
B. (2) only
C. (1) and (3) only
D. (1), (2) and (3)

A

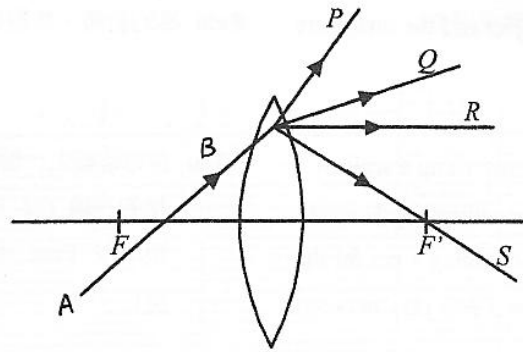
B

D

A

5. In the figure, F , F' are the foci of the convex lens and AB is an incident ray. Which of the following paths best represents the emergent ray?

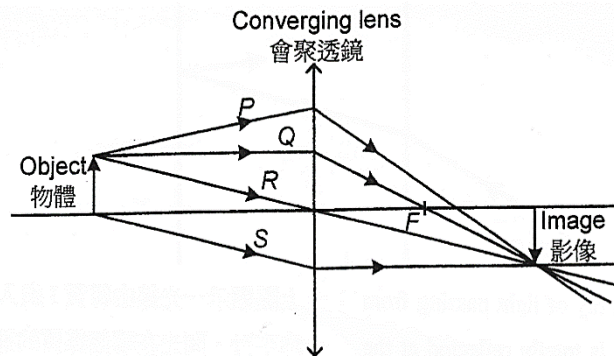
- A. P
B. Q
C. R
D. S



B

6. In the figure above, F is the focus of the converging lens. Which of the four rays is **INCORRECTLY** drawn?

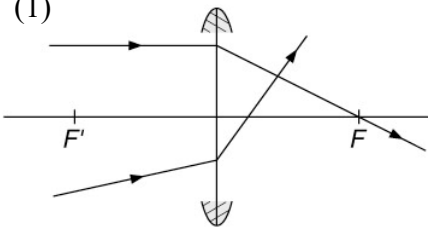
- A. P
B. Q
C. R
D. S



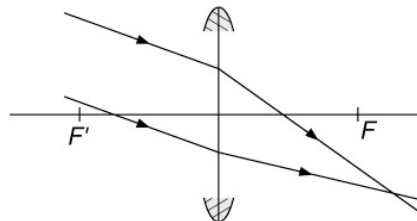
D

7. If F and F' are the foci of a convex lens, which of the following ray diagrams is/are **incorrect**?

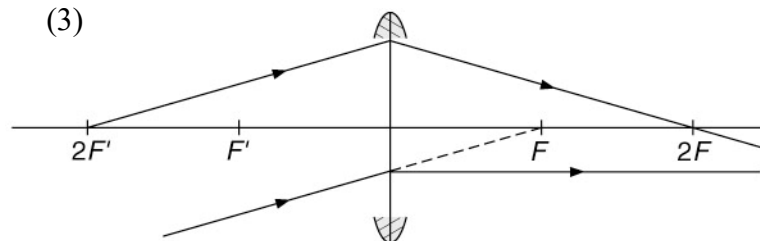
(1)



(2)



(3)

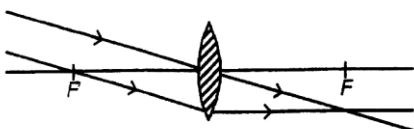


- A (3) only
B (1) and (2) only
C (2) and (3) only
D (1), (2) and (3)

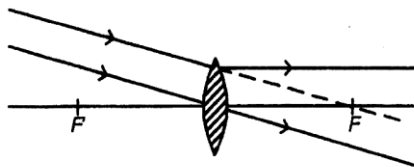
D

8. Which of the following ray diagrams is/are correct?

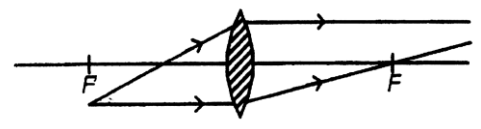
(1)



(2)



(3)

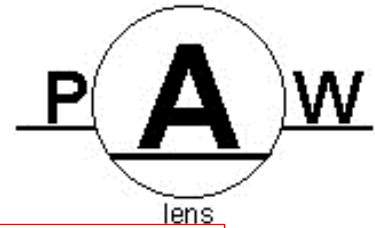


- A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

A

9. Some letters are viewed through a lens. The image of the letters is shown in the following figure. Which of the following statements is/are correct?

- (1) The letters are located within one focal length from the lens.
- (2) The image is virtual.
- (3) The lens is a convex lens.



- A (1) only
- B (3) only
- C (1) and (2) only
- D (1), (2) and (3)

(1): To produce a erect magnifying effect, the object must be placed within a distance of a focal length to the lens.
(2): Only a convex lens will produce a magnifying effect.

D

10. A distant object gradually approaches a convex lens along the principal axis. The image formed will
- (1) remain inverted throughout
 - (2) change from real to virtual after the object passes through the principal focus.
 - (3) increase in size before the object reaches the principal focus.

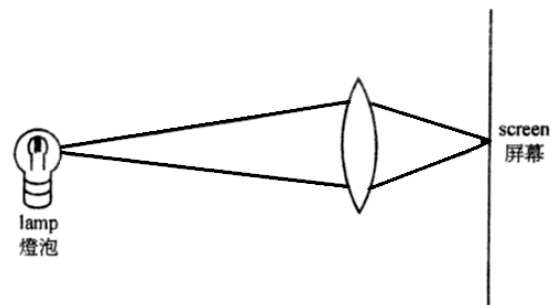
- A. (1) only
- B. (2) only
- C. (3) only
- D. (2) and (3) only

D

11. A convex lens L is placed between a screen and a lamp as shown in the diagram. A sharp image is formed on the screen. When the lens L is moved a certain distance towards the lamp, a sharp image is formed again on the screen. Which of the following statements concerning the second image is/are correct?

- (1) The image is larger than the first image.
- (2) The image is real.
- (3) The image is inverted.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (1), (2) and (3)

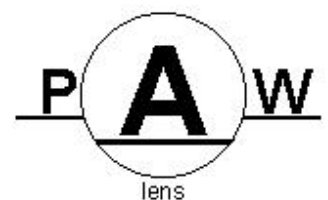


D

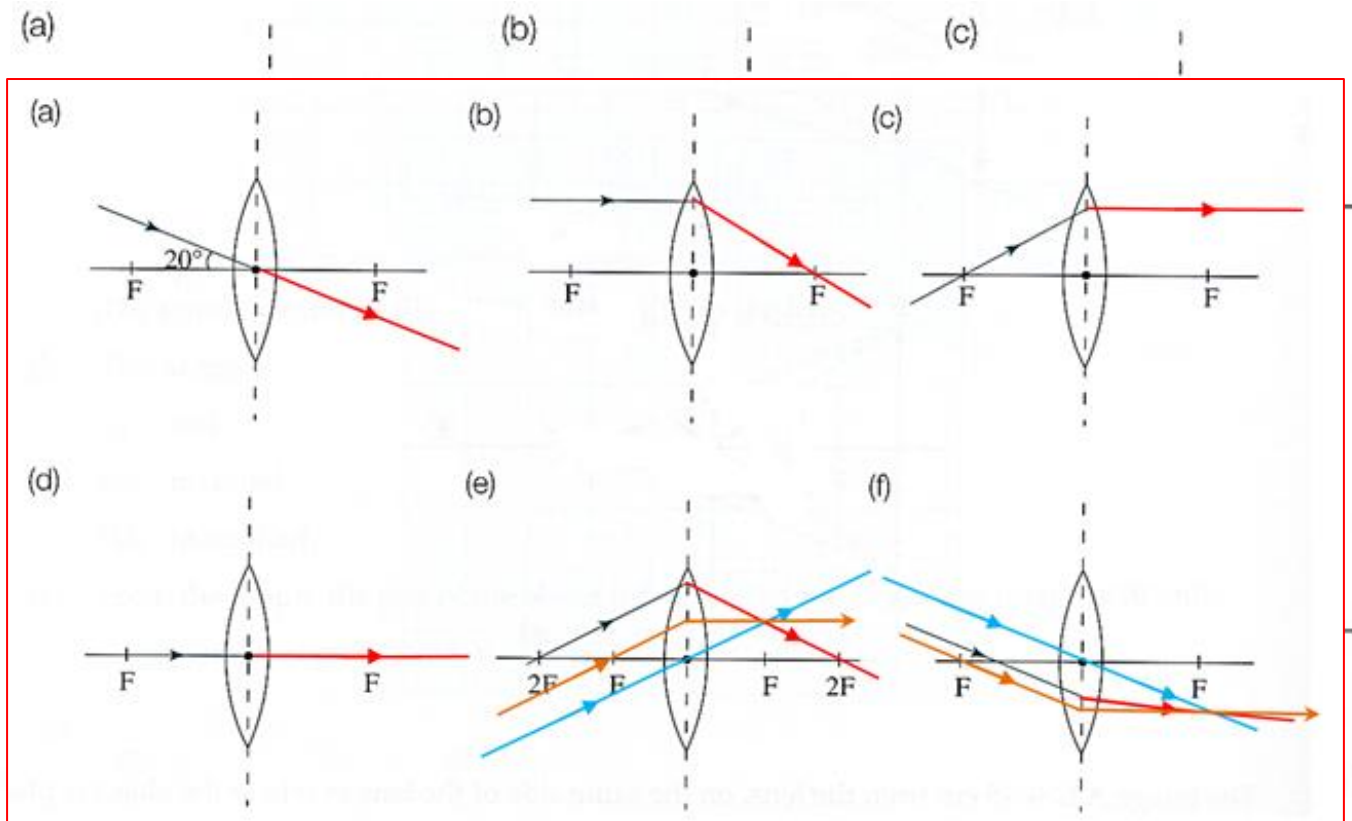
12. Some letters are viewed through a lens. The image of the letters is shown in the following figure. If the lens is moved away from the letters, which of the following correctly describes how the appearance of the image changes?

A

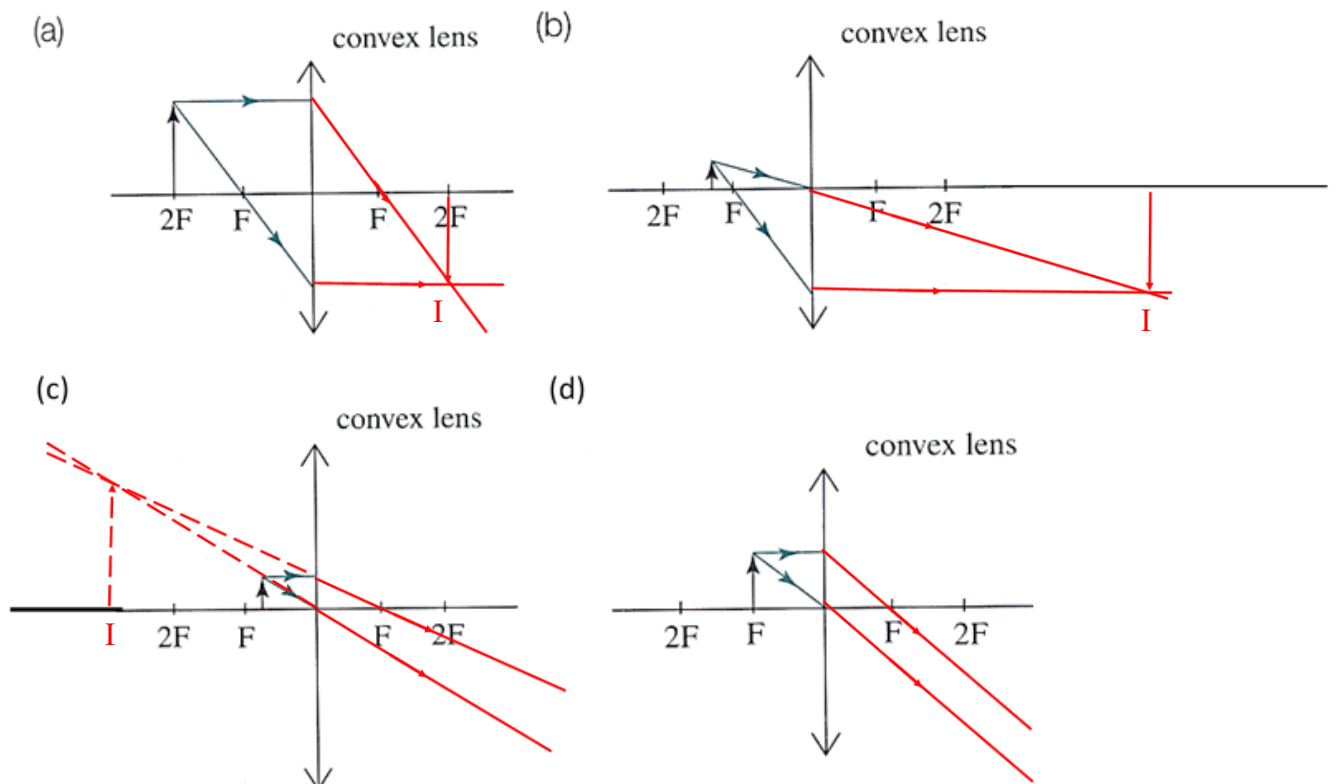
- A. Magnified, erect → magnified, inverted → diminished, inverted
- B. Magnified, erect → magnified, inverted → more magnified, inverted
- C. Magnified, erect → diminished, inverted → more diminished, inverted
- D. Magnified, erect → magnified, inverted → more magnified, erect



13. Draw the refracted rays.



14. Draw the refracted rays and locate the image.



15. Victor views a pearl through a lens as shown. His image can be seen through the lens.

- (a) (i) State the nature of his image.
(ii) What kind of lens is this?
- (b) What should be the distance between the lens and the pearl? Explain briefly.

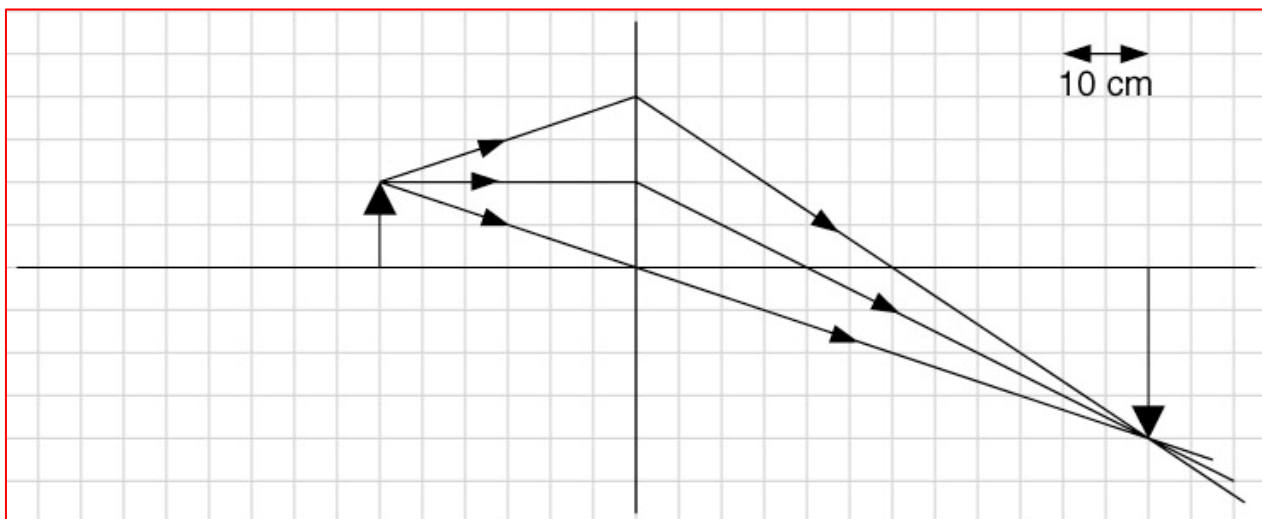


- (a) (i) virtual, erect, magnified
(ii) Convex lens
- (b) In order to produce a magnified image of the pearl, the distance between the pearl and the lens should be less than the focal length.

16. An object is placed in front of lens L. The following ray diagram shows that a light ray from the object is refracted by L.

- (a) What type of lens is L? Explain briefly.
- (b) By adding suitable light rays to the above diagram, find the image distance and the focal length of the lens.
- (c) State the nature of the image.

- (a) Convex lens because the ray converges.
- (b) Image distance = 60 cm
Focal length = 20 cm
- (c) Real, inverted, magnified



17. An object is placed in front of a converging lens as shown below. The four light rays shown come from the object.

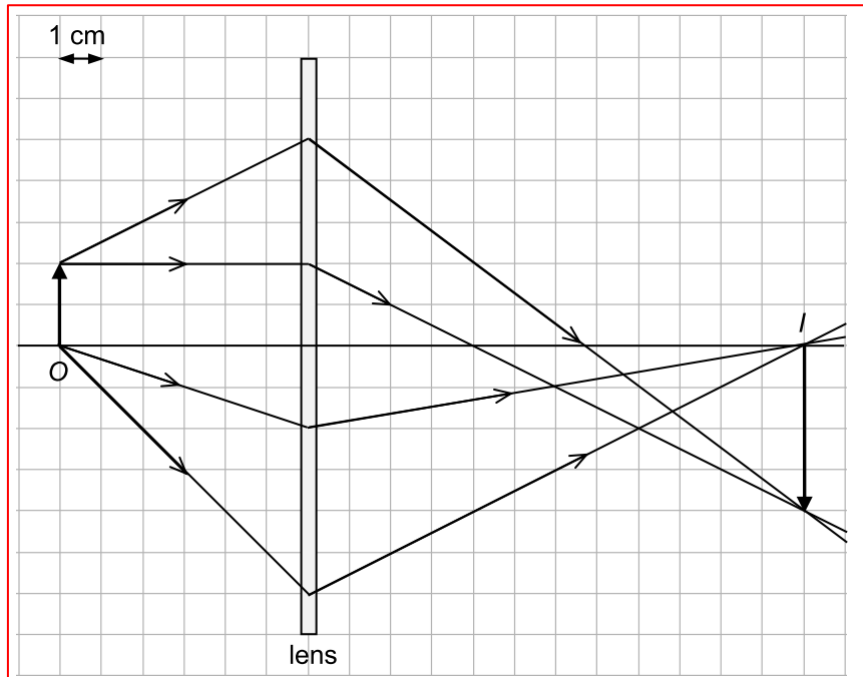
(a) (i) Complete the paths of the light rays in the figure and locate the image. (4 marks)

(ii) What is the focal length of the lens? (1 mark)

(b) Amy uses the lens to read a newspaper. A headline on the newspaper is at a distance of 1.5 cm from the lens. Can Amy use a screen to capture the image? Explain your answer. (2 marks)

(a) (ii) 4 cm

(b) No, because the image is virtual.



18. A letter 'F' illuminated by a lamp-housing is put in front of a convex lens as shown. The image of the letter is captured by a screen.

(a) Sketch the image formed on the screen. (1 mark)

(b) The lower half of the lens is then covered.

(i) Can the whole image be formed on the screen? Explain your answer with the aid of a ray diagram. (4 marks)

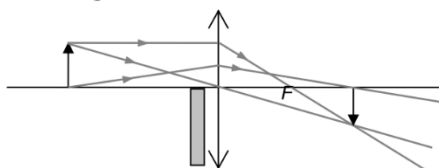
(ii) Does the brightness of the image change? Explain briefly. (2 marks)

(c) If the screen is moved a bit closer to the lens, the image will become blurry. Explain briefly. (1 mark)

(a)

(Correct image)

(b) (i) Yes, the whole image can be formed on the screen.



(Correct application of construction rules to locate the image)

(Ray from tail of object reaches tail of image via lens)

(ii) The image becomes dimmer as the amount of light forming the image decreases.

(c) Light from a point on the object will become a small patch instead of a point on the screen.

2 × 1A

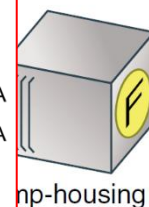
1A

1A

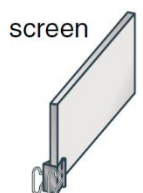
1A

1A

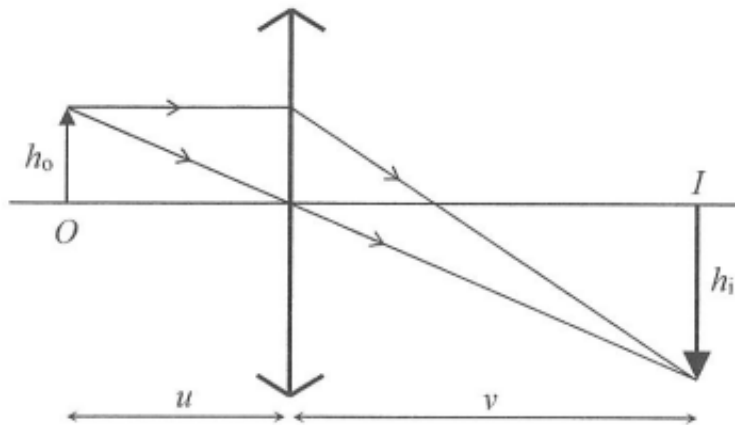
1A



convex lens



Magnification



Linear magnification (M)

$$M = \frac{v}{u} = \frac{h_i}{h_o}$$

v : image distance

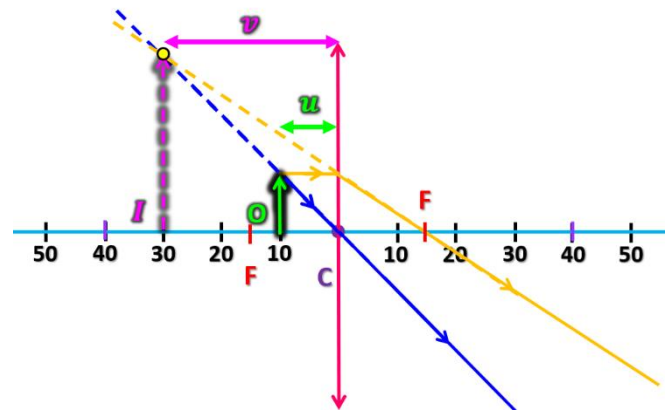
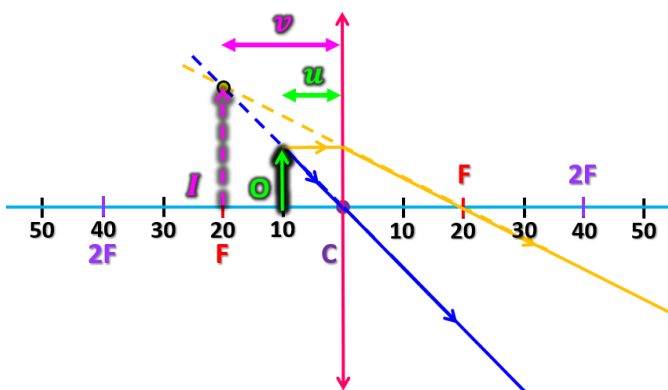
u : object distance

h_i : image height

h_o : object height

Magnification = $\frac{I}{O} = 2 = \frac{v}{u}$

Magnification = $\frac{I}{O} = 3 = \frac{v}{u}$



Lens formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

v : image distance (**real image: +ve, virtual image: -ve**)

u : object distance (**always +ve**)

f : focal length (**convex lens: +ve, concave lens: -ve**)

1. An object placed in front of a converging lens of focal length 10 cm forms an image 30 cm from the lens. If the image is real, what is the distance of the object from the lens?

- A. 7.5 cm
B. 8.5 cm
C. 12.5 cm
D. 15 cm

D
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
$$\Rightarrow \frac{1}{u} + \frac{1}{+30} = \frac{1}{+10}$$
$$\Rightarrow u = +15 \text{ cm}$$
 (Image distance is positive as the image is real.)

D

2. An object placed in front of a converging lens of focal length 10 cm forms an image 30 cm from the lens. If the image is virtual, what is the distance of the object from the lens?

- A. 7.5 cm
B. 8.5 cm
C. 12.5 cm
D. 15 cm

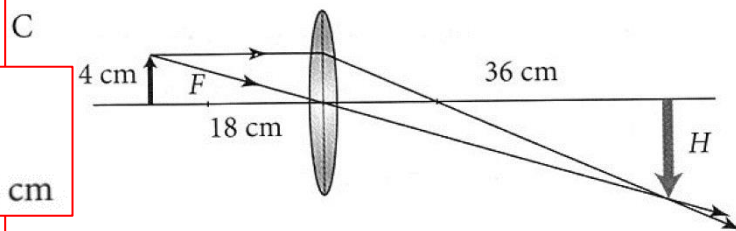
A
$$\frac{1}{u} + \frac{1}{-30} = \frac{1}{+10}$$
$$\Rightarrow u = +7.5 \text{ cm}$$
 (Image distance is negative as the image is virtual.)

A

3. An object of height 4 cm is placed 18 cm in front of convex lens of focal length 12 cm. A real image is formed 36 cm on the other side of the lens. What is the height of the image?

- A. 2 cm
B. 4 cm
C. 8 cm
D. 12 cm

$$\frac{H}{4} = \frac{36}{18}$$
$$\Rightarrow H = 8 \text{ cm}$$



C

4. An illuminated object placed 10 cm in front of a convex lens of focal length 8 cm forms a 4 times linearly magnified image on a screen. What is the distance between the object and the image?

- A. 32 cm
B. 40 cm
C. 48 cm
D. 50 cm

D Linear magnification = 4
$$\Rightarrow \frac{v}{u} = 4$$
$$\Rightarrow v = 4u = 4 \times 10 = 40 \text{ cm}$$
Distance between object and screen = $u + v = 50 \text{ cm}$

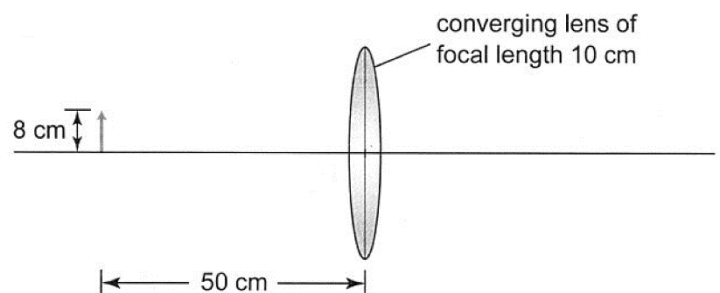
D

5. An object of height 8 cm is placed 50 cm from a converging lens of focal length 10 cm. What is the height of the image of the object?

- A. 2 cm
B. 3 cm
C. 4 cm
D. 5 cm

A
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
$$\Rightarrow \frac{1}{+50} + \frac{1}{v} = \frac{1}{+10}$$
$$\Rightarrow v = +12.5 \text{ cm}$$

$$m = \left| \frac{v}{u} \right| = \frac{12.5}{50} = 0.25$$
 (m is the linear magnification of the image.)
$$\Rightarrow \frac{H}{h} = 0.25$$
 (H and h are the heights of the image and the object respectively.)
$$\Rightarrow H = 0.25 \times 8 = 2 \text{ cm}$$



A

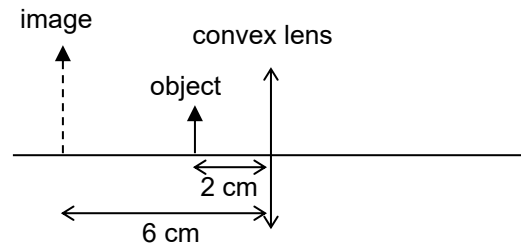
6. In the figure below, the positions of an object and its image formed by a convex lens are shown. The focal length of the lens is

- A. $\frac{1}{2} + \frac{1}{-6}$ cm
B. $\frac{(-6) \times 2}{-6+2}$ cm
C. $\frac{6 \times 2}{6+2}$ cm
D. $\frac{1}{2} + \frac{1}{6}$ cm

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1}{2} - \frac{1}{6}$$

$$f = 3 \text{ cm}$$



B

7. A student holds a magnifying glass 3 cm above a piece of paper on which the letters 'AT' is shown in the figure below. The magnification is 3. Find the image distance and the focal length of the lens.

Image distance Focal length of the lens

- A. 1 cm 0.75 cm
B. 1 cm 1.5 cm
C. 9 cm 2.25 cm
D. 9 cm 4.5 cm

$$m = \frac{v}{u}$$

$$3 = \frac{v}{3}$$

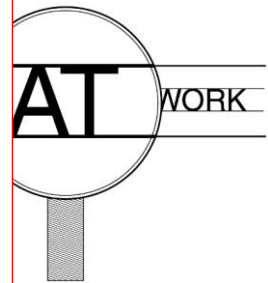
$$v = 9 \text{ cm}$$

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{3} - \frac{1}{9} = \frac{1}{f}$$

$$f = 4.5 \text{ cm}$$

the letters 'AT' is



D

8. The magnification of an image is 2, and the distance between the image and the lens is 8 cm. Which of the following may be the distance between the image and the object?

- (1) 4 cm
(2) 8 cm
(3) 12 cm
A (3) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

$$\text{By } m = \frac{v}{u}$$

$$u = \frac{v}{m} = \frac{8}{2} = 4 \text{ cm}$$

Since the image is magnified, the lens must be convex. There are two scenarios the image can be formed.

Scenario 1:

If the object distance is between f and $2f$, the image and the object are on opposite sides of the lens.

Distance between the image and the object = $4 + 8 = 12 \text{ cm}$

Scenario 2:

If the object distance is smaller than f , the image and the object are on the same side of the lens.

Distance between the image and the object = $8 - 4 = 4 \text{ cm}$

$\therefore$ (1) and (3) are correct. (2) is incorrect.

B

9. An object is placed 20 cm in front of a converging lens of focal length 30 cm. Which of the following statements about its images is/are correct (1):

- (1) The image is real.
(2) The image is magnified.
(3) The image is erect.
A. (1) only
B. (1) and (2)
C. (2) and (3)
D. (1), (2) and (3)

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{20} + \frac{1}{v} = \frac{1}{30}$$

$$v = -60$$

Thus, the image is virtual.

(2): The magnification of the object is

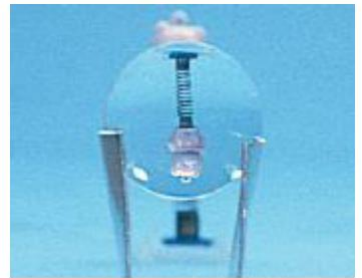
$$= \frac{60}{20} = 3$$

(3): Since the object distance is less than 1 focal length, the image is a magnifying and erect one.

C

10. A toy is viewed through a convex lens as shown.

- State the nature of the image.
- If the object and image distances are 10 cm and 5 cm respectively, find the magnification of the image.



(a) Real, inverted, diminished

$$\begin{aligned}\text{(b) Magnification} &= \frac{v}{u} \\ &= \frac{5}{10} \\ &= 0.5\end{aligned}$$

11. In the figure below, a woman holds a lens in front of her right eye.

- What type of lens does the woman hold? Explain your answer.
- Suppose the distance between the lens and the woman's right eye is 6 cm and the focal length of the lens is 12 cm. Find the image distance.

(a) Since the image is magnified, the lens is a convex lens.

(b) Applying the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{6} + \frac{1}{v} = \frac{1}{12}$$

$$v = -12$$

$\therefore$ The image distance is 12 cm.



12. A folding magnifier with a magnification of 4 is shown below. It consists of a single lens supported by a stand of height 9 cm. The object to be viewed is placed below a square opening.

- State the nature of the image formed (erect or inverted, real or virtual).
- Find the image distance and the focal length of the lens.

(a) The image is erect and virtual.

(b) By $m = \frac{v}{u}$,

$$\text{image distance } v = mu = 4 \times 9 = 36 \text{ cm}$$

Apply the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{9} + \frac{1}{-36} = \frac{1}{f}$$

$$f = 12$$

$\therefore$ The focal length of the lens is 12 cm.



販賣,違者必究

13. An object is placed 12 cm from a convex lens. A clear image of the object is form on a screen 5 cm away from the lens.

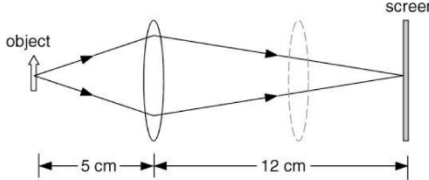
(a) Find the magnification of the image. (2 marks)

(b) (i) Suppose the object and the screen are fixed while the lens is moved towards the object. How far should the lens be moved so that a sharp image can be caught on the screen again? Explain your answer briefly. (3 marks)

(ii) Find the magnification of this new image. (1 mark)

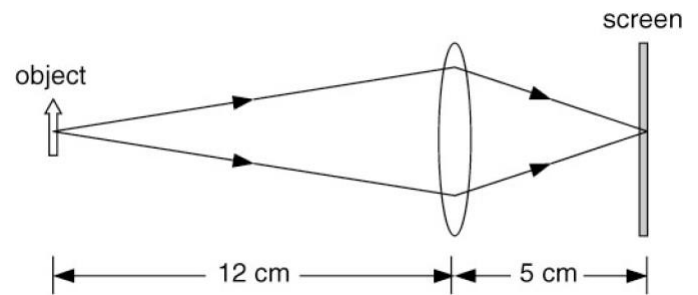
(a) Magnification = $\frac{\text{image distance}}{\text{object distance}}$
 $= \frac{5}{12}$
 $= 0.417$

(b) (i) If we view the screen through the lens, its image will be formed at the position of the object.
 Therefore, when the object distance is 5 cm, the image distance will be 12 cm.



Distance moved by the lens = $12 - 5 = 7$ cm

(ii) Magnification = $\frac{\text{image distance}}{\text{object distance}} = \frac{12}{5} = 2.4$



14. In the figure below, an illuminated letter 'L' is placed 30 cm in front of a convex lens and its image formed by the lens is caught by a translucent screen. The image is half as large as the object.

(a) (i) Is the image real or virtual? Explain your answer.

(ii) Sketch the shape of the image seen by the observer.

(b) Find the image distance and the focal length of the lens.

(a) (i) The image is real since it can be captured by a screen.

(ii) (Correct image)



(b) By $m = \frac{v}{u}$,

$$\text{image distance } v = mu = \frac{1}{2} \times 30 = 15 \text{ cm}$$

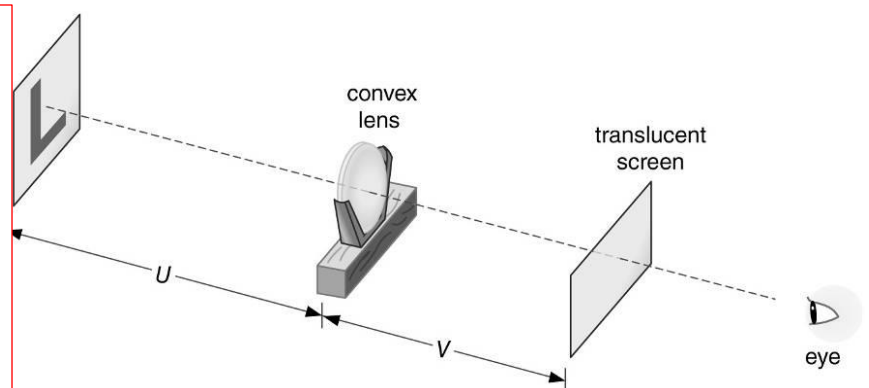
Applying the lens formula,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$\frac{1}{30} + \frac{1}{15} = \frac{1}{f}$$

$$f = 10$$

$\therefore$ The focal length of the lens is 10 cm.



15. Anthony uses the set-up shown in Figure a to study the image formation of a convex lens. He puts an illuminated letter F in front of the lens and an image is formed on a translucent screen. He finds that the object distance u is larger than the image distance v .

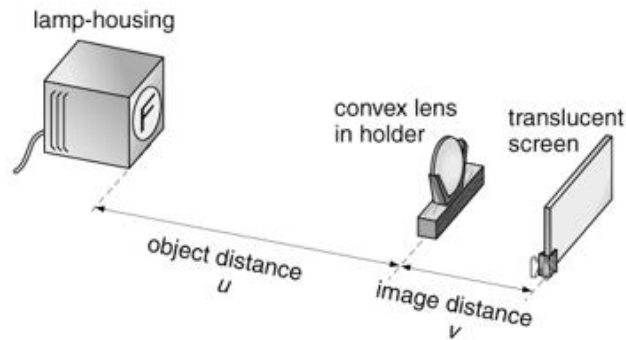


Fig a

- Sketch the image on the screen as observed from the right side of the set-up.
- Is the image real or virtual? Explain your answer.
- Anthony changes the object distance u and measures the values of v . After obtaining several sets of data, he constructs the graph shown in Figure b.
 - Find the magnification of the image when $u = 20$ cm.
 - If the object is moved closer to the lens, how will the image size change?
 - Find the value of u when $u = v$. Hence find the focal length of the lens.
- What will happen to the image if the upper half of the lens is covered by a piece of opaque paper?
- Explain, with the help of a ray diagram, why an image cannot be formed on the screen when the object distance is shorter than the focal length of the lens.

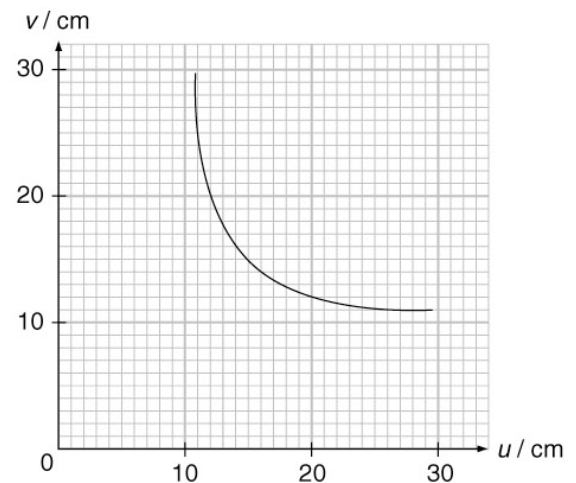


Fig b

(Dotted lines to show the virtual property)↗

16. An illuminated object is placed in front of a lens and a sharp image is formed on a screen on the other side of the lens. The magnification of the image is 4. The focal length of the lens is 20 cm.

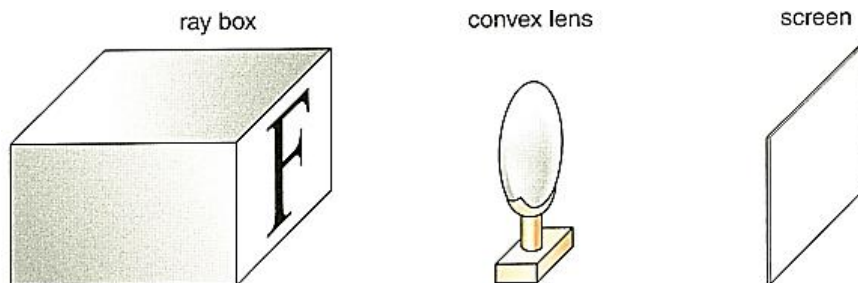
- (a) What type of lens is used? Explain briefly.
(b) Find the object and the image distances.

(a) The lens is a convex lens
since the image is real.

(b) By $m = \frac{v}{u}$,
 $v = mu = 4u$
Put $v = 4u$ and $f = 20$ into the lens formula,
$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$
$$\frac{1}{u} + \frac{1}{4u} = \frac{1}{20}$$
$$u = 25$$

$\therefore$ The object distance is 25 cm.
Image distance = $mu = 4 \times 25 = 100$ cm

17. The figure below shows the set-up of an experiment. An image of a letter 'F' is formed on a screen by a convex lens.



- (a) Is the image real or virtual? (1 mark)
(b) If the convex lens is equidistant from the ray box and the screen, what is the size of the image relative to the object? (1 mark)
(c) If the set-up is placed underwater, how will the image distance change? Explain briefly. (4 marks)

(a) Real
(b) Same size as object
(c) The image distance increases.
This is because the refractive index of water is higher than that of air.
Light is refracted by a smaller amount when it passes the water-lens interface and so the focal length of the lens increases.